

Verifier-Gated LLM NetOps: A Preregistered, Artifact-Backed Evaluation of Input-Sanitization and Deterministic Verification Against Adversarial Intent

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Abstract—We report a preregistered paired confirmatory experiment (study_id cand-2026-03) that measured whether template-based input sanitization combined with deterministic verifier-gating (and at-most-one automated repair) changes the rate at which a deterministic validator accepts LLM-generated CLI configurations under adversarially mutated intents. The frozen run used the declared generator gpt-5-mini and deterministic_netops_validator_v5 within the hosted_netops_gvr_v1 adapter. Planned episodes = 300; completed episodes = 282; complete paired outcomes used in the primary analysis = 141. Observed per-pair acceptance: Baseline 135/141 (95.7447%); Guarded 141/141 (100.0%); paired absolute difference = 0.04255319148936165; exact two-sided McNemar $p = 0.03125$; 95% bootstrap percentile CI (10,000 resamples, seed = 1729) = [0.014184397163120567, 0.07801418439716312]. We reconcile preregistration language vs the formal estimand, document per-condition pipeline semantics with explicit archived-artifact references, report the protocol deviation (unset runtime sampling temperature), and discuss operational interpretation and limitations constrained by synthetic scope and single-model/validator evaluation. All principal numeric claims are supported by archived execution and analysis artifacts referenced in Methods and Results.

I. INTRODUCTION

Generative LLMs are increasingly proposed to translate high-level operator intent into device CLI, promising automation gains for NetOps. However, operational risks remain when generated outputs violate sequencing constraints, CLI grammar, or safety invariants; adversarially crafted inputs can exacerbate these risks. Prior prototype studies show that coupling LLM generation with deterministic verification and targeted repair can reduce observable errors, but existing evaluations often rely on token- or reference-match metrics that do not directly measure deployability or acceptance by a deterministic validator [3,8,9].

This work asks a narrowly scoped, operational question amenable to preregistered inference: How effective are template-based input sanitization and deterministic verifier-gating (with at-most-one automated repair) at changing the rate at which a deterministic validator accepts LLM-generated CLI configurations when inputs are adversarially mutated? The study cand-2026-03 preregistered a single formal estimand (paired_success_rate_difference_guarded_minus_baseline) and a confirmatory paired test. Crucially, because the

registered generation semantics were shared_initial_candidate, the paired comparison isolates downstream guarded-pipeline effects (sanitization, gating, permitted repair, and final validation) applied to a single shared LLM draft per intent rather than differences in initial generation.

Contributions. (1) A preregistered, artifact-backed paired experiment executed end-to-end and reconciled against the archived preregistration; (2) a precise, artifact-grounded description of per-condition pipeline semantics that demonstrates shared_initial_candidate reuse; (3) quantitative confirmatory results showing a small but statistically detectable absolute change in deterministic-validator acceptance under the guarded pipeline on a synthetic adversarial-intent bank; and (4) an explicit reconciliation of preregistration natural-language hypothesis wording with the formal estimand and a disclosure of a protocol deviation (unset sampling temperature) together with reproducibility guidance.

II. RELATED WORK

Deterministic network-verification tools are a practical foundation for validator-gated NetOps pipelines because they check forwarding and configuration invariants at the configuration or header-space level and can run fast enough for interactive uses. Classic header-space analyses and their derivatives provide reachability and invariant-checking semantics over packet headers and device-wide rule sets, while streaming/online tools (e.g., VeriFlow/NetPlumber) focus on real-time policy checking as configurations change [W1882012874,W158224344,W2122695394]. These verifiers are valuable for gating because they give binary deployability decisions (accept/reject) against explicit formal properties; their effectiveness, however, depends on the granularity and correctness of the formal specification (what is asserted as a safety invariant) and on whether the verifier models dynamic protocol-level behavior or only static configuration-level constraints.

Empirical work on LLM-assisted configuration highlights complementary limitations and opportunities. Evaluations such as NetConfEval and other recent NetOps-focused studies report that LLMs frequently produce syntactic, ordering, or semantic-sequence errors when tasked with intent-to-CLI

translation; generate-validate-repair architectures—where an LLM draft is checked by a deterministic tool and then repaired—have been demonstrated in prototype settings to reduce observable errors compared to unconstrained generation [W4399629579,W4388629193]. Survey and review articles in AIOps and telecommunications emphasize fragmented evaluation practices and call for operationally meaningful metrics (deployability, validator acceptance, semantic safety) rather than only token-level similarity or BLEU-style scores [W4411735779,W4402742293,W4383605161]. Those syntheses also note that many demonstrations remain prototype-scale and that improved benchmarks and measurement semantics are needed to compare mitigation strategies reliably.

On the security and orchestration side, prompt-injection and indirect-prompt attacks against LLM-integrated applications have been documented and motivate defenses such as input-sanitization, policy gating, and stricter tool-mediated checks in operational pipelines [W438886073,W4407163436]. Concurrent work on tool-augmented LLM patterns (e.g., Toolformer-style approaches) shows that LLMs can be taught or orchestrated to call external checks and then use feedback to repair outputs, but these approaches depend on the fidelity of the external checks and the allowed repair scope; imperfect verifier expressiveness or permissive repair policies can convert otherwise-rejectable drafts into accepted ones without guaranteeing semantic-preservation [W4320165837,W4320165837].

Positioning. Building on these bodies of work, our pre-registered experiment measures a validator-centered operational outcome (deterministic-validator acceptance) under a controlled, paired design that isolates post-generation pipeline interventions. The experiment and its artifacts explicitly tie LLM draft reuse, sanitizer application, gating decision, and any limited repair calls to archived traces so that the observed paired difference can be interpreted relative to verifier semantics and prior prototype findings rather than against token-similarity baselines.

III. METHODOLOGY

Overview and preregistration. The study was preregistered as cand-2026-03 before observing outcomes (preregistration_sha256: ba6ed25e84c7240f94f2d50b3e778fbfffd810ff2db632df8946e85f0f1373). The preregistration specifies a paired confirmatory design (shared_initial_candidate generation semantics), a single primary estimand (paired_success_rate_difference_guarded_minus_baseline), and concrete analysis procedures (exact two-sided McNemar for the paired comparison; bootstrap-percentile 95% CIs with 10,000 resamples and seed = 1729). These analysis parameters and bootstrap metadata are archived (analysis/analysis_log.jsonl).

Task bank, adversary model, and scope. Tasks were produced by deterministic_netops_task_generator_v5; each task encodes: (i) an initial device-state snapshot, (ii) an operator intent template (intent-to-configuration), and (iii) a required command sequence and per-field constraints (e.g., must-shutdown-before-MTU-change). The synthetic bank exercises

admin, MTU, and VLAN updates and a small set of workflow policies (protected interfaces, group-activity minima, must-be-down constraints). The adversary model used deterministic mutation heuristics applied to these templates (string-level mutations, slot-overrides, token insertions) as archived in execution/task_manifest.jsonl (see representative entries such as execution/task_manifest.jsonl -> task-000001 payload). Scope was intentionally limited to these synthetic templates and the deterministic CLI grammar encoded in the validator; no production device logs or vendor-specific private data were used.

Orchestration, declared models, and protocol deviation. The orchestration adapter was hosted_netops_gvr_v1 and the declared generator was gpt-5-mini (execution/model_configuration.json). The deterministic validator was deterministic_netops_validator_v5; validator outputs include explicit validation_before and validation_after objects with parsed command counts, touched-field counts, enumerated violation_codes, and final_state snapshots (examples: execution/scoring/task-000001-baseline.json and execution/scoring/task-000001-guarded.json). The preregistered pipeline constrained generation semantics to shared_initial_candidate (independent_condition_generation = false). The archived model_configuration.json records declared_model_version = "gpt-5-mini" and temperature = null; we disclose this unset temperature as a protocol deviation because provider-side defaults may influence sampling. To preserve internal comparability despite the unset parameter the pipeline used a single shared generation per pair (evidence: analysis/deterministic_reconciliation.json: stage_counts.shared_initial_generation = 150) and provider-call accounting (hosted_model_call_event_count = 156, completed = 147, failed = 9) corroborates the actual call pattern.

Per-pair timeline and concrete artifact evidence (shared_initial_candidate semantics). The registered shared_initial_candidate semantics produced the following deterministic per-pair timeline; each step and its archived artifact(s) are listed to permit exact forensic inspection: 1) Task instantiation and adversarial mutation: recorded in execution/task_manifest.jsonl (task payload contains the sanitized-orig intent and any slot changes). Representative entry: task-000001 payload (execution/model_manifest and execution/task_manifest.jsonl). 2) Shared initial generation (single LLM call per pair): output saved to execution/responses/task-<id>-shared-initial.txt (e.g., execution/responses/task-000001-shared-initial.txt). Provider-stage accounting records shared_initial_generation = 150 (analysis/deterministic_reconciliation.json). 3) Baseline branch scoring: deterministic_netops_validator_v5 evaluates the shared_initial_candidate; baseline validator trace is in execution/scoring/task-<id>-baseline.json (validation_before and validation_after objects show parsed_command_count, required_command_count, valid flag, and violation codes when present). 4) Guarded branch pre-processing and gating: a template-based input-sanitizer is applied (sanitizer

logic and results are recorded in the task payload fields). The sanitized candidate is presented to the verifier-gate; if the sanitized candidate fails gating the policy allows at-most-one automated repair call. Repair-stage provider-call traces for episodes invoking repair are present under execution/provider_calls and referenced by execution/scoring artifacts (repair_provider_trace_path fields). The pipeline-level repair budget (≤ 1 repair call per episode) is enforced by the orchestration adapter (preregistration and execution manifests). Evidence: stage_counts.repair = 6 and repair traces linked in execution/scoring files and in analysis/deterministic_reconciliation.json. 5) Final guarded validation: the validator re-evaluates the sanitized or repaired candidate, storing validation_after and final_configuration in execution/scoring/task-`<id>`-guarded.json. Paired artifacts for the same shared_initial_candidate include the shared_initial_candidate_sha256 value present in both baseline and guarded scoring artifacts (e.g., execution/scoring/task-000001-baseline.json and execution/scoring/task-000001-guarded.json share the same shared_initial_candidate_sha256).

Sanitizer and gating specifics (artifact-grounded). The input-sanitizer is template-driven: it enforces required slots, normalizes numeric fields (e.g., MTU integer normalization), and rejects or rewrites fields that violate per-task allowed_fields/allowed_touched_fields constraints; sanitized outputs are recorded in the task payload (execution/task_manifest.jsonl task_payload.constraints and task_payload.intent). The gating policy is deterministic: if deterministic_netops_validator_v5 reports violations that fall into a policy-rejection set (e.g., protected-interface modification or missing required-step violations), the candidate is rejected; if violations are repairable under permitted-scope rules, the orchestrator may invoke a single repair generation attempt. These behaviors and the task-specific policies are visible in the task payload constraints and in per-episode validator traces (execution/scoring/* files).

Repair semantics and auditability. Repairs are restricted by preregistered policy: one additional LLM call per episode, limited to localized edits (slot corrections, sequence re-ordering) rather than free-form regeneration. Repair calls and resulting candidates are fully auditable: provider traces, shared_initial_candidate_sha256, repair_prompt_sha256, and repair_provider_trace_path appear in execution/scoring artifacts for repaired episodes (representative repaired episodes are traceable and correspond to the six repair-stage counts in analysis/deterministic_reconciliation.json). The six discordant n10 cases observed in the paired contingency table correspond to episodes where a permitted repair converted an initially rejected draft into a validator-accepted final candidate; individual scoring artifacts show validation_before (invalid) and validation_after (valid) snapshots.

Outcome definitions, missingness handling, and exclusions. For each adversarial intent (paired unit) the binary outcome equals 1 if the deterministic validator would accept the final candidate for deployment and 0 otherwise. The preregistered missingness_plan excluded a paired unit only when both

paired runs were technical failures (model API failure, verifier timeout, or parser error). Execution accounting: planned_episode_count = 300, completed_episode_count = 282, failed_episode_count = 18, complete_pairs = 141 (analysis/missingness_summary.csv). Both-missing pairs = 9; provider-call audit: hosted_model_call_event_count = 156, completed = 147, failed = 9 (analysis/deterministic_reconciliation.json). All exclusions and failure reasons are archived in execution/execution_log.jsonl and analysis/missingness_summary.csv.

Statistical-analysis procedures (artifact-backed). The preregistered primary test is the exact two-sided McNemar test on paired binary outcomes; contingency counts used by the test are archived at analysis/paired_contingency_table.csv (n11 = 135, n10 = 6, n01 = 0, n00 = 0). Paired effect estimation is presented as the absolute difference in success rates (guarded - baseline) with a bootstrap-percentile 95% CI obtained from 10,000 resamples (bootstrap_seed = 1729); bootstrap parameters and logs are archived in analysis/analysis_log.jsonl. Secondary analyses (exploratory) were planned for subgroup effects by mutation class (string-level, token-injection, slot-override) and for a benign-control false-positive-blocking rate; the benign-control (M = 50) was not executed in this run and therefore no empirical false-positive-blocking rates are reported.

Determinism, provenance, and reproducibility notes. All run artifacts required to reproduce the analysis are archived and referenced above (execution/* and analysis/*). The single protocol deviation (temperature = null in execution/model_configuration.json) is disclosed; because the pipeline used shared_initial_candidate semantics and provider-call accounting shows a single shared generation per pair (stage_counts.shared_initial_generation = 150), paired comparability is preserved for this run. For deterministic reproduction we recommend explicitly setting temperature = 0 and re-running the archived execution/task_manifest.jsonl with identical orchestration to produce bit-identical provider traces under the same provider semantics. Representative artifact paths for forensic verification: execution/task_manifest.jsonl, execution/responses/task-000001-shared-initial.txt, execution/scoring/task-000001-baseline.json, execution/scoring/task-000001-guarded.json, analysis/deterministic_reconciliation.json, analysis/paired_contingency_table.csv, analysis/analysis_log.jsonl.

Pre-registered safeguards and contamination controls. The preregistration included contamination controls (artifact hashing and embargo rules for hazardous artifacts) and explicit exclusion rules for sanitized benign intents with excessive slot-drop (> 0.3). Contamination summaries and integrity checks are archived in analysis/contamination_summary.csv and the execution/execution_manifest.json. Any artifact-flagging and masking procedures applied during archival are recorded; flagged hazardous raw artifacts are embargoed per the pre-registered contamination policy.

Summary of methodological guarantees and limitations. Methodologically, the design enforces a paired comparison

isolating downstream guarded-pipeline effects by reusing a single shared_initial_candidate per pair and limiting repairs to a single bounded attempt. Auditability is provided by per-episode validator traces, provider-call traces, and deterministic reconciliation artifacts. Limitations inherent in the design (single-model, synthetic tasks, validator coverage, and the protocol deviation noted above) are described in Threats to Validity and Limitations; these do not alter the integrity of the reported artifact-grounded paired comparison but bound external generalization.

IV. RESULTS

Execution summary and primary counts. The archived execution manifest reports `planned_episodes = 300`, `completed_episodes = 282`, `failed_episodes = 18` and `model_calls_used = 156` (`execution/execution_manifest.json`). After applying the preregistered paired exclusion rule (exclude pair only when both paired runs are technical failures) the confirmatory analysis used `complete_pairs = 141` (`analysis/missingness_summary.csv`). Condition-level totals used in the primary estimand are recorded in `analysis/condition_summary.csv` and reproduced here: `baseline_successes = 135`, `baseline_failures = 6` (observed = 141), `guarded_successes = 141`, `guarded_failures = 0` (observed = 141). The condition summary artifact path: `analysis/condition_summary.csv`.

Paired contingency, test, and interval. The paired contingency table archived at `analysis/paired_contingency_table.csv` is `n11 = 135` (`guarded=1`, `baseline=1`), `n10 = 6` (`guarded=1`, `baseline=0`), `n01 = 0` (`guarded=0`, `baseline=1`), `n00 = 0` (`guarded=0`, `baseline=0`). Using the preregistered exact two-sided McNemar test on these paired outcomes yields `p = 0.03125` (confirmatory result stored in `analysis/results.json` and summarized in `analysis/paired_contingency_table.csv`). The observed paired point estimate (`guarded - baseline`) = 0.04255319148936165. The 95% bootstrap percentile confidence interval (method and parameters preregistered; `bootstrap_resamples = 10,000`; `bootstrap_seed = 1729`) is [0.014184397163120567, 0.07801418439716312] (see `analysis/analysis_log.jsonl` and `analysis/paired_difference_figure.svg` for bootstrap metadata and visualization). These exact numeric artifacts are archived at `analysis/paired_contingency_table.csv` and `analysis/analysis_log.jsonl`.

Discordant-case diagnostics and repair-stage association. All six discordant pairs (n10) are traceable in the archive. Provider-call accounting in `analysis/deterministic_reconciliation.json` records `stage_counts`: `shared_initial_generation = 150` and `repair = 6`; `hosted_model_call_event_count = 156` (`completed = 147`, `failed = 9`). The one-to-one correspondence between the repair-stage call count and n10 suggests the discordant conversions were associated with episodes where the guarded pipeline either applied sanitization that altered the candidate prior to gating or invoked the single allowed automated repair call. Per-episode scoring artifacts for

discordant pairs contain validator before/after snapshots (`validation_before` and `validation_after`) and, when present, repair-related provider-trace references (example artifact group: `execution/scoring/task-00000X-baseline.json` and `execution/scoring/task-00000X-guarded.json`). Representative artifacts demonstrating the shared_initial_candidate reuse and validator snapshots include `execution/responses/task-000001-shared-initial.txt`, `execution/scoring/task-000001-baseline.json`, and `execution/scoring/task-000001-guarded.json`.

Mechanism observed in archived traces. The archived validator traces show two mechanistic pathways that yielded n10 discordances: (A) template-based sanitizer canonicalized or repaired malformed slots so the sanitized candidate passed the validator when the raw shared_initial_candidate had previously failed; or (B) the guarded policy permitted one automated repair LLM call whose output, after re-validation, satisfied the validator. The archive documents both sanitizer activity (task payloads in `execution/task_manifest.jsonl`) and repair-stage call traces where present (provider traces referenced from scoring artifacts). The per-episode scoring artifacts include `validation_before` and `validation_after` objects which record the validator’s `normalized_configuration`, `violation_count`, and `validator_id = deterministic_netops_validator_v5`; these fields permit exact forensic comparison of pre- and post-guarded candidates.

Missingness sensitivity. Nine paired units were excluded because both runs were technical failures (`analysis/missingness_summary.csv`: `both_missing_pairs = 9`). Applying the conservative sensitivity that treats excluded pairs as failures yields `baseline_acceptance = 135/150 = 0.9000` and `guarded_acceptance = 141/150 = 0.9400` (reported in Methods and briefly summarized here as a conservative bound). The archived missingness artifact path is `analysis/missingness_summary.csv`; the reconciliation artifact that records completed/failed provider calls and stage counts is `analysis/deterministic_reconciliation.json`.

Provider-call audit and reproducibility-relevant details. Provider-call accounting (`analysis/deterministic_reconciliation.json`) shows `hosted_model_call_event_count = 156`, `completed_hosted_model_call_event_count = 147`, `failed_hosted_model_call_event_count = 9`, `cache_hit_count = 0`, `cache_miss_count = 147`, and `stage_counts`: `shared_initial_generation = 150`, `repair = 6`. The model configuration artifact (`execution/model_configuration.json`) records `declared_model_version = "gpt-5-mini"` and `temperature = null`; this unset temperature value is a protocol deviation discussed in Methods and Disclosure. The provider-call audit plus the shared_initial_generation count provide artifact-grounded evidence that the per-pair generation semantics used a single shared initial candidate (see `execution/responses/*-shared-initial.txt`) and that at most one repair-stage call occurred for the small subset of episodes that needed it. Paths: `analysis/deterministic_reconciliation.json` and `execution/model_configuration.json`.

Representative per-task artifact inspection. For

reproducibility and reader verification we point to representative per-task artifact groups archived in execution/ (examples listed in the evidence bundle):

- task-000001: execution/responses/task-000001-shared-initial.txt, execution/scoring/task-000001-baseline.json, execution/scoring/task-000001-guarded.json;
- task-000002..task-000006: analogous shared-initial, baseline, and guarded response and scoring artifacts (see execution/responses/ and execution/scoring/). These representative artifacts show identical shared_initial_candidate content referenced by both baseline and guarded scoring records along with validator validation_before/validation_after snapshots and the scorer_id = deterministic_netops_validator_v5, enabling exact verification of the reported paired outcomes.

Unexecuted preregistered items and limitations relevant to results. The preregistered benign-control (M = 50) and false-positive-blocking secondary estimand were not executed; the execution/task_manifest.jsonl and responses directories contain no benign-control artifacts. Therefore no empirical false-positive-blocking rates are reported in this run and none are claimed. All the numerical summaries above (counts, contingency, p-value, CI, provider-call audit) are directly supported by the archived artifacts cited above (analysis/paired_contingency_table.csv; analysis/condition_summary.csv; analysis/missingness_summary.csv; analysis/deterministic_reconciliation.json; execution/responses/*-shared-initial.txt; execution/scoring/*-baseline.json).

V. DISCUSSION

Clarifying the preregistration versus the formal estimand. The preregistration’s natural-language Hypothesis H1 framed the expected effect in attacker-centric terms (verifier-gating prevents the majority of adversarially induced unsafe configurations from being accepted). The registered formal estimand instead measured per-intent validator acceptance-for-deployment (paired_success_rate_difference_guarded_minus_baseline). Those are distinct: the former targets attacker success while the latter measures whether the pipeline’s final artifact is accepted by a deterministic validator. The archived results show an increase in validator acceptance under the guarded pipeline (baseline 135/141 → guarded 141/141; paired difference $\approx +0.04255$; exact McNemar $p = 0.03125$; analysis/condition_summary.csv, analysis/paired_contingency_table.csv). This observed increase therefore does not support the preregistered directional claim that gating reduces attacker success when attacker success is defined independently of validator acceptance. Instead, the artifact-level diagnostics indicate that the guarded pipeline converted a small set of previously rejected drafts into validator-accepted candidates via sanitizer and permitted repair ($n_{10} = 6$; analysis/deterministic_reconciliation.json). Consequently, readers should interpret the confirmatory result strictly with respect to the archived formal estimand (validator

acceptance) and not equate acceptance-improvement with a demonstrated reduction in attacker success absent an independent semantic-safety oracle.

Mechanism and artifact-grounded diagnostics. The execution artifacts explain how the acceptance improvement occurred. Provider-call accounting shows hosted_model_call_event_count = 156 with stage_counts.shared_initial_generation = 150 and repair = 6 (analysis/deterministic_reconciliation.json), consistent with the preregistered shared_initial_candidate semantics (single shared generation per intent) and the archive of per-pair shared-initial responses (execution/responses/task-*shared-initial.txt). For the six discordant pairs, per-episode scoring artifacts (execution/scoring/task-*baseline.json and execution/scoring/task-*guarded.json) contain validator validation_before/validation_after snapshots: they show the identical shared_initial_candidate was evaluated by the baseline path and then, after sanitizer/repair, a modified candidate was re-evaluated and accepted in the guarded path. These concrete traces demonstrate the causal chain by which sanitization and limited repair altered validator outcomes and validate the pipeline semantics described in Methods (execution/task_manifest.jsonl and scoring artifacts are cross-referenced in the archive).

Operational safety trade-offs and policy implications. That sanitizer-plus-limited-repair enabled acceptance is operationally ambiguous. If repairs are semantics-preserving corrections (e.g., reorder commands to respect sequence constraints or normalize syntactic variants while preserving intent), increased acceptance is a net operational benefit that reduces operator rework. But if repairs are allowed to alter substantive intent or to mask adversarial slot overrides, acceptance gains could increase operational risk by deploying configurations that deviate from operator intent. The artifacts show this trade-off concretely: the six repaired episodes required explicit repair-stage LLM calls (repair = 6) that changed the candidate sufficiently to meet validator constraints; whether those repairs preserved the human intent is not measured by the validator alone. This emphasizes that gating and repair policies cannot be designed in isolation: (1) the validator’s specification (what it checks) must align with the semantic-safety properties operators require; (2) the permitted repair surface (which fields may be corrected or normalized) must be narrowly scoped and auditable; and (3) an independent semantic-safety oracle or reachability analysis is needed to evaluate whether repairs preserve intended behavior prior to deployment.

Reproducibility, protocol deviation, and paired comparability. The archived model_configuration.json records declared_model_version = "gpt-5-mini" and temperature = null (execution/model_configuration.json). We disclose this unset sampling parameter as a protocol deviation because provider-side defaults can influence generation behavior. Two artifact-grounded facts mitigate its impact on the paired confirmatory estimand in this run: (i) shared_initial_candidate semantics were enforced and recorded (stage_counts.shared_initial_generation

= 150), producing a single initial sample per intent that both Baseline and Guarded paths evaluated (execution/responses/*-shared-initial.txt; execution/scoring/*-baseline.json and *-guarded.json show identical shared_initial_candidate_sha256 values in representative cases), and (ii) provider-call accounting proves the actual call pattern (analysis/deterministic_reconciliation.json). Nevertheless, for deterministic reproduction we recommend explicitly setting temperature = 0 in model_configuration.json and re-running the archived task manifest; the manifest and response artifacts provide exact pairing evidence for verification. Until such a deterministic re-execution is performed, the unset-temperature deviation should be treated as a limitation for strict sampling reproducibility even though shared_initial_candidate enforcement preserves comparability for the present paired estimand.

Sample, missingness, and interpretive caution. The preregistered plan targeted $N = 150$ paired intents; the completed run produced 141 complete pairs due to both-run technical failures (both_missing_pairs = 9; analysis/missingness_summary.csv). Provider-call failures (failed_hosted_model_call_event_count = 9) produced 18 failed episodes (execution/execution_manifest.json). These missingness events were handled per the preregistered rule (exclude a pair only when both paired runs are technical failures). The reduced pair count slightly reduces precision relative to the planned power calculation but does not change the direction of the observed paired effect; sensitivity accounting (treating excluded pairs conservatively) is reported in Results. Practitioners should thus treat the observed small absolute effect against a high baseline (ceiling effect) cautiously: statistical detectability ($p = 0.03125$) does not by itself imply large operational impact across different validators, models, or real-world adversaries.

Design and measurement recommendations grounded in artifacts. Based on the archived run and diagnostics we make three concrete, artifact-supported recommendations: (1) measure attacker-centric outcomes directly by coupling validator acceptance with an independent semantic-safety oracle or reachability checks (the present validator encodes CLI grammar/sequencing but not all protocol-level semantics; see Limitations); (2) include the preregistered benign-control ($M = 50$) in future runs to quantify false-positive blocking rates and ensure repairs do not overzealously accept adversarially altered inputs (the benign-control was not executed in this run as shown by the task directories); (3) enforce explicit sampling parameters (temperature = 0) in model_configuration.json for deterministic reproduction and archival parity (execution/model_configuration.json currently records temperature = null). Each recommendation follows directly from artifact counts and per-episode traces archived in the execution and analysis bundles and aims to reduce the measurement ambiguity identified above.

VI. LIMITATIONS

- Synthetic task bank and intent templates — not real production traffic or operator logs; results may not generalize to field datasets.
- Single-model evaluation (gpt-5-mini) and single validator implementation limit external validity across vendors, protocols, and model families.
- Validator coverage constrained to encoded safety properties (CLI grammar, required sequences, protected-interface invariants); some protocol-level or dynamic runtime semantics are not represented.
- Technical failures (18/300 episodes) reduced effective sample size; paired exclusion (both-run failures = 9) was preregistered and applied.
- Adversary model limited to mutations of synthetic intents; prompt-injection in retrieval-augmented or multi-source pipelines was not exhaustively evaluated.
- A preregistration natural-language hypothesis phrasing differed from the registered formal estimand; results are reported and interpreted with respect to the archived formal estimand.
- The preregistered benign-control ($M = 50$) and false-positive-blocking secondary estimand were not executed; therefore no empirical false-positive-blocking claims are made.

VII. CONCLUSION

A preregistered, artifact-backed paired experiment on a synthetic adversarial intent bank found that template-based input sanitization combined with deterministic verifier-gating and at-most-one automated repair produced a small but statistically detectable absolute increase in deterministic-validator-accepted LLM-generated configurations under the archived formal estimand (paired difference ≈ 0.04255 ; $p = 0.03125$; 95% CI [0.01418, 0.07801]). Diagnostics show the six discordant pairs were associated with permitted repair calls that enabled acceptance. Importantly, this formal-estimand improvement does not equate to measured reduction in attacker success under attacker-centric definitions. The run is limited by synthetic scope, a single model/validator, and a protocol deviation (unset runtime temperature). Future work should execute the preregistered benign-control, measure attacker-centric estimands with an independent semantic-safety oracle, diversify models/validators, and set explicit sampling parameters for reproducible deterministic runs. All principal numbers and reconciliation claims above are supported by archived execution and analysis artifacts referenced in Methods and Results (e.g., execution/task_manifest.json; execution/responses/task-000001-shared-initial.txt; execution/scoring/task-000001-baseline.json; execution/scoring/task-000001-guarded.json; analysis/paired_contingency_table.csv; analysis/condition_summary.csv).

DISCLOSURE STATEMENT

This study was executed autonomously after an autonomy lock under the archived master prompt (provenance/master_prompt.txt, SHA-256: 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15ba). Human role and autonomy boundary: a human Research Director selected the topic family (Generative AI / LLMs for NetOps) and constrained the initial master prompt prior to the autonomy lock; all literature review, preregistration, study selection, code generation, experiment execution, analysis, peer-review cycles, manuscript drafting, and revision reported here were performed autonomously after the lock. Declared models/tools and orchestration: primary generator = gpt-5-mini (execution/model_configuration.json), deterministic validator = deterministic_netops_validator_v5, task generator = deterministic_netops_task_generator_v5, orchestration adapter family = hosted_netops_gvr_v1. Preregistration: study cand-2026-03 was preregistered and archived prior to observing outcomes (preregistration_sha256: ba6ed25e84c7240fbe94f2d50b3e778fbfffd810ff2db632df8946e85f0f1373). Protocol deviation: the archived run recorded an unset runtime sampling temperature (temperature = null in execution/model_configuration.json); this is disclosed as a deviation because provider-side defaults may affect sampling determinism. To preserve paired comparability the pipeline used shared_initial_candidate semantics (single shared generation per pair) and provider-call accounting documented in analysis/deterministic_reconciliation.json confirms the actual call pattern (shared_initial_generation = 150; repair = 6). Key archived artifact references (examples): execution/task_manifest.jsonl; execution/responses/task-000001-shared-initial.txt; execution/scoring/task-000001-baseline.json; execution/scoring/task-000001-guarded.json; analysis/deterministic_reconciliation.json; analysis/paired_contingency_table.csv; analysis/condition_summary.csv; analysis/analysis_log.jsonl; execution/model_configuration.json; execution/execution_manifest.json. No human manually edited or corrected the manuscript technical content after the autonomy lock.

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